

CS6493: NATURAL LANGUAGE PROCESSING

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Natural Language Processing

Subject Code

CS - Computer Science

Course Number

6493

Academic Unit

Computer Science (CS)

College/School

College of Computing (CC)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

CS5481 Data Engineering or
CS5487 Machine Learning: Principles and Practice or
CS5489 Machine Learning: Algorithms and Applications or
CS5491 Artificial Intelligence or
SDSC5001 Statistical Machine Learning I or
SDSC6001 Statistical Machine Learning II or
SDSC8007 Deep Learning

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

This course introduces algorithms and techniques for natural language processing, from computational linguistics for text processing to information extraction for language understanding. The topics include statistical and neural based language modeling, word representation, pretrained language models such as BERT and large language models, such as GPT and Llama. Basic and advanced natural language processing tasks, such as machine translation dialog systems question answering, text classification/labeling/tagging, and knowledge graph will also be introduced.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if DEC-A1 app.)		DEC-A2	DEC-A3
1	Perform statistical and neural processing for syntactic analysis of text properties.			x	
2	Perform information extraction for understanding of text semantics.			x	
3	Assess the effectiveness of natural language processing and understanding for real-world problems.		x		
4	Apply techniques in natural language processing and understanding for innovative applications.		x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lecture	Students will engage with the algorithms and techniques for natural language processing and understanding. Related real-world applications such as machine translation and dialog systems will be introduced.	1, 2, 3	2 hours/ week

2	Tutorial	Students will work on a different problem set each week during the tutorial sessions, through which they can discover the main characteristics of different natural language processing techniques and integrate them for real-world problems. They will also be invited to present their solutions, and the class will be encouraged to provide comments.	1, 2, 3	1 hour/ week
3	Assignments	Students will implement selected natural language processing and understanding approaches, apply these approaches to real-world problems and interpret the results. In this way, students can analyse the performance of different approaches.	1, 2, 3	
4	Project	Students will create a new system design and implement appropriate natural language processing approaches for innovative applications. The students will apply the principles they have learnt from the course for their design.	4	
5	Examination	Examination will include questions to assess the capability of students 1) to identify the important features of natural language processing and understanding approaches; 2) to perform critical evaluation of different algorithms for real-world language problems; 3) to modify or design algorithms for challenging applications.	1, 2, 3	

Assessment Tasks / Activities (ATs)

	ATs	CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Assignments	1, 2, 3	30	Expect to have two assignments with programming elements for algorithm implementation.	Yes
2	Project	4	30	Around 1-3 students in a group to finish the project	Yes

Continuous Assessment (%)

60

Examination (%)

40

Examination Duration (Hours)

2

Minimum Examination Passing Requirement (%)

30

Additional Information for ATs

For a student to pass the course, at least 30% of the maximum mark for the examination must be obtained.

Assessment Rubrics (AR)**Assessment Task**

Assignments (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The ability to implement and assess the effectiveness of different algorithms and techniques.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Project (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The ability and creativity in designing and implementing appropriate algorithms and techniques for innovative applications.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

The extent to which the students can understand the algorithms and techniques, apply them with appropriate modification or design new solutions for different applications, and evaluate their performances.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Assignments (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The ability to implement and assess the effectiveness of different algorithms and techniques.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

Marginal

(B-, C+, C) Moderate to Basic

Failure

(F) Not even reaching marginal level

Assessment Task

Project (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The ability and creativity in designing and implementing appropriate algorithms and techniques for innovative applications.

Excellent

(A+, A, A-) High

Good

(B+, B) Significant

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Assessment Task

Examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The extent to which the students can understand the algorithms and techniques, apply them with appropriate modification or design new solutions for different applications, and evaluate their performances.

Excellent

(A+, A, A-) High

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Failure

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Part III Other Information

Keyword Syllabus

NLP basics: language models, word representations (distributional representation and contextualized representation) recurrent neural networks, attention mechanism, transformers, pretrained language models and large language models.

NLP task examples: machine translation, dialog systems, question answering, text classification/tagging/ labeling, knowledge graph, and other advanced topics.

Reading List

Compulsory Readings

Title	
1	Daniel Jurafsky and James H. Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, 3rd edition (online), 2020.
2	Christopher D. Manning, Hinrich Schutze, Foundations of Statistical Natural Language Processing, MIT Press, 1999.
3	Jacob Eisenstein, Natural Language Processing, online, 2018.

Additional Readings

Title	
1	Nil